

Spatial Panel Econometrics: Evaluating Weather Variables and Solar Yields Across Zambia (2020 to 2025)

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1. Overview

As Zambia diversifies its energy mix away from heavy dependence on hydropower, solar photovoltaic (PV) capacity has become a central pillar of the nation's energy security strategy. While solar irradiance (GHI) serves as the primary physical driver of photovoltaic output, ambient temperature introduces a secondary, compounding dynamic: PV cells experience a loss in conversion efficiency as operating temperature rises above the standard test condition of 25°C. This study evaluates that dual dynamic using a spatial panel spanning all ten provinces of Zambia, quantifying the magnitude of the thermal penalty and its behavioural variation across diverse regional microclimates.

2. Data and Econometric Framework

2.1 Dataset Scale and Provenance

Spatial Panel Dimensions. A balanced 21,920 observation spatial panel ($n = 10$ provinces, $T = 2,192$ daily observations per province) tracking daily weather metrics and simulated energy outputs across Zambia from 2020 to 2025.

Data Sources. Daily surface global horizontal irradiance (GHI, measured in kWh/m²/day) and ambient temperature (measured in °C) series were extracted from the Open Meteo Historical Weather API, which compiles ERA5 reanalysis data generated by the European Centre for Medium Range Weather Forecasts (ECMWF).

Simulation Mechanics. Daily energy yield (expressed in kWh/m²/day) was simulated from observed weather series using a fixed baseline panel efficiency rating, a linear thermal derating penalty applied for temperatures exceeding 25°C, and an additive stochastic noise term. Because energy yield is simulated rather than metered from physical inverters, results function strictly as a methodological validation framework and a quantification of the engineered thermal penalty, rather than calibrated engineering output figures.

2.2 Econometric Specification

To control for unobserved, time invariant regional heterogeneity, such as permanent topographical differences or baseline provincial microclimates, the analysis implements a One-Way Panel Fixed Effects (Within Effects) model:

$$Yield_{it} = \alpha_i + \beta_1(GHI_{it}) + \beta_2(Temperature_{it}) + \varepsilon_{it}$$

Here α_i absorbs province specific baseline intercepts, isolating pure within province temporal variation over time. To account for serial correlation and heteroscedasticity common in daily weather data, standard errors are cluster robust at the province level.

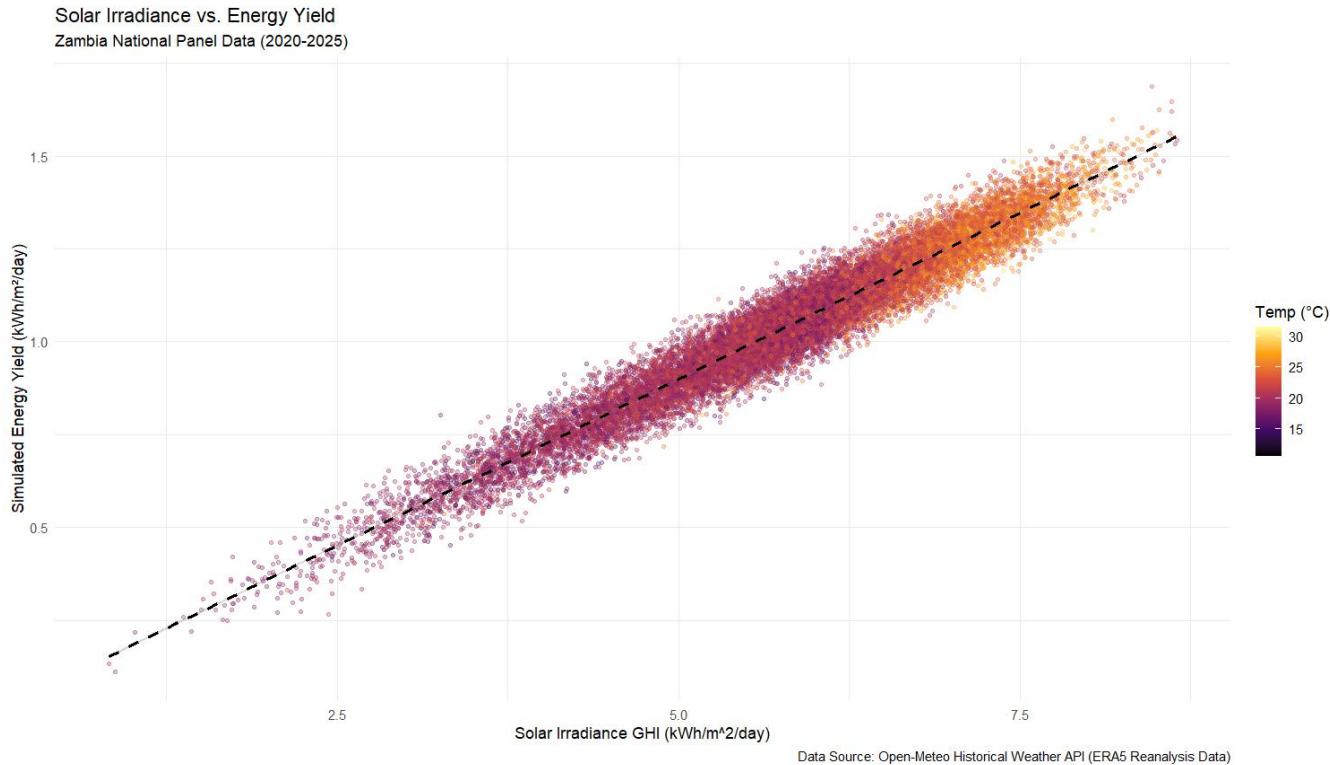


Figure 1. Solar irradiance (GHI) versus simulated energy yield, Zambia national panel data, 2020 to 2025, $n = 21,920$. Point color encodes ambient temperature. Source: Open Meteo Historical Weather API (ERA5 Reanalysis Data).

3. Econometric Findings and Statistical Diagnostics

3.1 Regression Estimates

The fixed effects estimation yields stable, statistically robust parameters.

Solar Irradiance ($\beta_1 = 0.18009$, $t = 804.54$, $p < 2.2e-16$) confirms a strong positive baseline response where each additional unit of daily irradiance ($\text{kWh}/\text{m}^2/\text{day}$) scales power generation directly, holding regional fixed effects constant.

Ambient Temperature ($\beta_2 = -0.00079$, $t = -4.62$, $p = 3.77e-06$) isolates a statistically significant negative coefficient, capturing the thermal degradation penalty on photovoltaic performance.

Simulated Energy Yield	
(kWh/m ² /day)	
Solar Irradiance GHI	0.18009 *** (0.000224)
Temperature	−0.00079 ***

(0.000171)

Province Fixed Effects	Yes
Clustered SE (province)	Yes
Observations	21,920
R ²	0.9388
Adjusted R ²	0.93878
F Statistic	168,062 ***

*** $p < 0.001$. Standard errors, shown in parentheses, are clustered at the province level.

Model Fit. The model accounts for an R squared of 0.9388 (adjusted R squared = 0.93878), with an overall F statistic of 168,062 ($p < 2.2\text{e-}16$). Comparing these parameters against an initial pooled OLS specification shows virtually zero coefficient drift, confirming that unobserved regional differences did not distort the slope parameters.

3.2 Statistical Diagnostics

Multicollinearity. Evaluated via the Variance Inflation Factor (VIF), returning a clean score of 1.40, well below standard concern thresholds, ensuring independent interpretation of irradiance and temperature.

Residual Analysis. Diagnostic checks confirm linearity and homoscedasticity, with cluster robust standard errors correcting for temporal autocorrelation across the daily panel series.

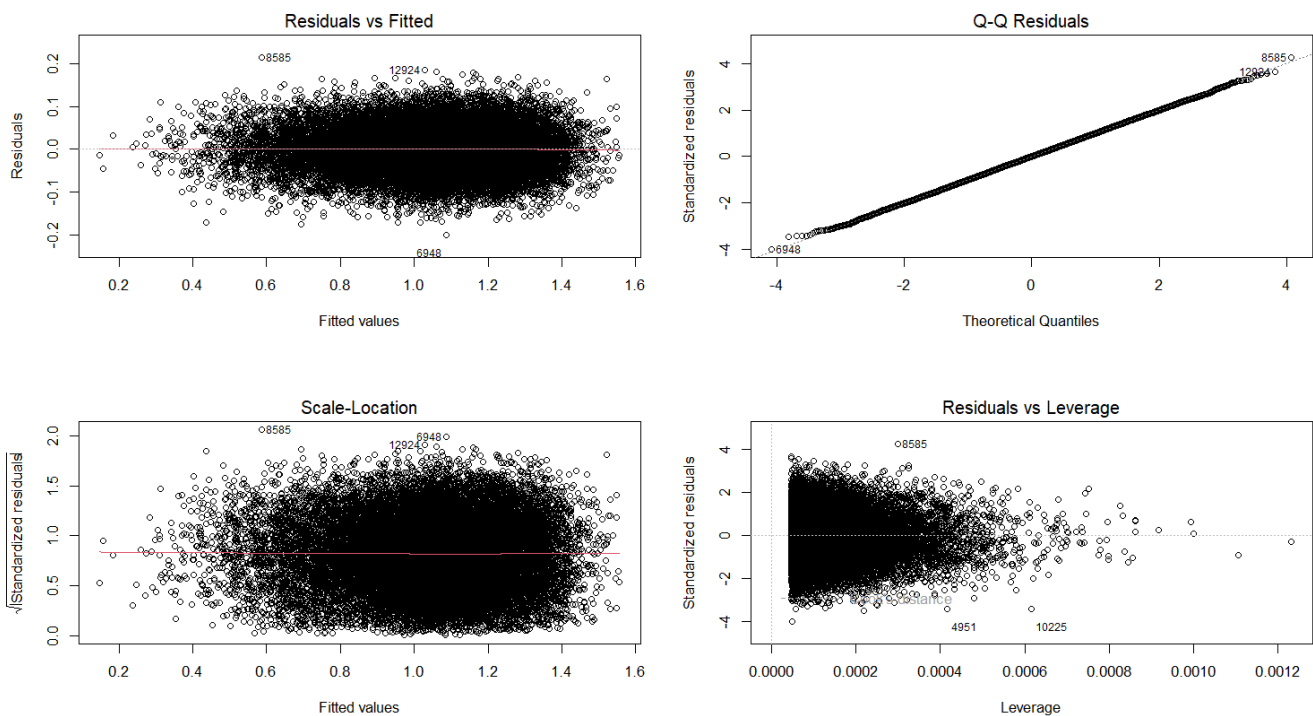


Figure 2. Residual diagnostic plots underlying the model: Residuals vs Fitted, Normal Q Q, Scale Location, and Residuals vs Leverage.

4. Regional Variations and Spatial Insights

Implementing a panel fixed effects framework ensures that estimates are driven strictly by within province variation over time rather than cross regional baseline distortions. Low lying valley zones and high altitude plateau provinces experience divergent local microclimates. Because the thermal derating penalty binds above a fixed 25°C threshold, provinces enduring higher ambient temperature baselines experience more frequent efficiency contractions. Consequently, two geographic sites with identical annual irradiance totals can yield different realized capacity factors once local temperature regimes are factored in, a critical consideration for capital allocation and project siting.

5. Limitations and Future Work

Simulated Metrics. Because the dependent variable is synthetic, future extensions should calibrate model coefficients against real world inverter telemetry data.

Model Extensions. Future iterations can incorporate nonlinear polynomial or interaction terms to test whether thermal derating accelerates during extreme irradiance spikes, alongside environmental factors such as cloud cover, relative humidity, and dust or soiling losses.

6. Conclusion

Integrating both GHI and ambient temperature into a panel fixed effects framework avoids the structural oversimplifications inherent in single variable weather assessments. While irradiance remains the dominant driver of photovoltaic yield across Zambia, temperature imposes a statistically robust efficiency penalty that materializes on hot, high irradiance days. This framework establishes a rigorous, transparent foundation for advanced energy planning and site selection.
